CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Hydroponics Flow Lab

Goal: Model a gravity-fed hydroponic path that distributes water evenly without leaking or flooding.

35 minutes | 8 steps | Engineering and Design | Playbook A meeting 23

Ask | Plan | Build | Test | Improve

You are done when

Show the measured flow result and explain the one change that made distribution more even.

You will use

- Clean tubing
- Cups
- Reclaimed channels
- Water trays
- Measuring cups
- Towels
- Plant-site models
- Teacher-controlled tubing cutter
- Wet mats
- Sink bucket

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Place the dry tray on the wet mat.
- Measure the test water.

2 Step 2: Wet tool safety — 4 min

- Keep water inside the tray.
- Wipe each spill with a towel.
- Do not cut tubing yourself.
- Keep electronics away from the wet zone.

3 Step 3: Plan the flow — 3 min

- Draw the water cup.
- Draw the water path.
- Mark each plant spot and the end cup.

4 Step 4: Build the dry model — 7 min

- Mark the cut line on the tube.
- Give the tube to the teacher.
- Wait for the cut, then connect the tube path.
- Check every connection while dry.

5 Step 5: Run the water — 6 min

- Pour the measured water into the water cup.
- Watch each plant spot.
- Measure the collected water.

6 Step 6: Change the flow — 6 min

- Change one slope, hole, or loose connection.
- Repeat with the same water amount.
- Record the new collected amount.

7 Step 7: Dry the lab

4 min

- Drain the tray into the sink bucket.
- Place wet parts on the drying tray.
- Dry the table and floor.

8 Step 8: Exit check

2 min

- Write the collected amount.
- Name the flow change.

What to turn in

A measured flow result and one change that made the flow more even.

No computer? Another way to do it

A dry path model with one teacher pour demo.

Finished early? Next challenge

Flow Alarm: Design a visible marker that shows when one plant site receives too much or too little water.

Stay safe

Keep water away from outlets and electronics, wipe spills immediately, and use no pump unless the teacher controls it.

Grade targets

Grade 5: Measure water at each plant site with a cup. Mark more or less.
Grade 6: Calculate the percent of water recovered and name where it was lost.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.